

Healthcare Data Analysis

Data Cleaning

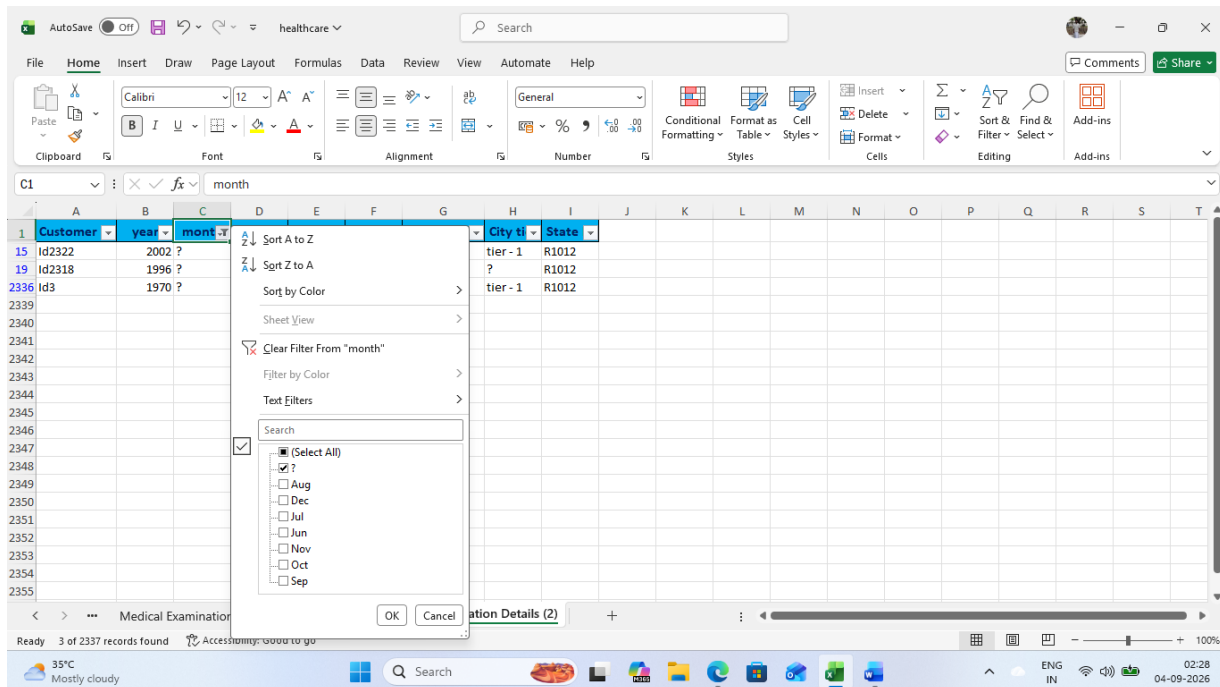
Data Cleaning – Question 1

Question: Check for the number of missing values marked with '?' in each column of the "Medical Examinations" table and "Hospitalization Details" table.

Explanation: This means identify and count every '?' entry in each column of the two named tables before continuing with the analysis.

Technique used: Excel Filter / column filtering to display '?' values. The supplied screenshot shows the filter being used.

Formula used: Not shown as a formula in the source screenshot; the method is Filter.



Data Cleaning – Question 2

Question: Fill in the missing values of 'month' with Sep and 'year' with its average, rounded to the nearest integer.

Explanation: This means enter Sep wherever Month is missing, then calculate the average Year, round it to a whole number, and use that value for the missing Year cells.

Technique used: Filter/identify missing cells, manually enter Sep, and calculate the average Year with ROUND and AVERAGE.

Formula used: Exact formula visible in the source screenshot: =ROUND(AVERAGE(B2:B2338),0)

Microsoft Excel interface showing a table with the following data:

Customer	year	month	date	children	charge	Hospital tier	City tier	State
Id2322	2002	Sep	19	0	750	tier - 3	tier - 1	R1012
Id2318	1996	Sep	18	0	770.38	tier - 3	?	R1012
Id3	1970	Sep	11	3	60021.4	tier - 1	tier - 1	R1012

Microsoft Excel interface showing a table with the following data:

Customer	year	month	date	children	charge	Hospital tier	City tier	State
Id18	1963	Aug	5	1	47896.79	tier - 1	tier - 3	R1024
Id17	1958	Nov	16	1	47928.03	tier - 2	tier - 3	R1011
Id16	1962	Aug	27	0	48173.36	tier - 1	tier - 3	R1011
Id15	1961	Dec	21	1	48517.56	tier - 1	tier - 3	R1024
Id14	1968	Dec	1	3	48549.18	tier - 1	tier - 3	R1016
Id13	1962	Oct	11	0	48673.56	tier - 1	tier - 2	R1013
Id12	1965	Oct	27	1	48675.52	tier - 1	tier - 2	R1013
Id11	1959	Jul	22	0	48824.45	tier - 2	tier - 1	R1011
Id10	1978	Dec	29	0	48885.14	tier - 1	tier - 2	R1013
Id9	1963	Sep	4	1	48970.25	tier - 1	tier - 2	R1013
Id8	1958	Jun	27	2	49577.66	tier - 2	tier - 2	R1013
Id7	1994	Oct	27	1	51194.56	tier - 1	tier - 3	R1011
Id6	1962	Aug	4	0	52590.83	tier - 1	tier - 3	R1011
Id5	1989	Jun	19	0	55135.4	tier - 1	tier - 2	R1012
Id4	1991	Jun	6	1	58571.07	tier - 1	tier - 3	R1024
Id3	1970	Sep	11	3	60021.4	tier - 1	tier - 1	R1012
Id2	1977	Jun	8	0	62592.87	tier - 2	tier - 3	R1013
Id1	1968	Oct	12	0	63770.43	tier - 1	tier - 3	R1013

Data Cleaning – Question 3

Question: Determine the most frequently occurring values in the 'smoker', 'hospital tier', and 'city tier' columns, and fill in the missing values accordingly.

Explanation: This means find the mode (most frequently occurring value) separately for Smoker, Hospital Tier and City Tier, then use those values to fill the missing records.

Technique used: MODE/MATCH helper technique and Find/Replace are shown in the source screenshots.

Formula used: Exact mode formula visible for Smoker: =MODE(MATCH(\$H\$2:\$H\$2336,\$H\$2:\$H\$2336,0)). The screenshots also show MATCH being used as a helper. No separate exact Hospital Tier/City Tier formula is clearly visible, so no formula is invented here.

The screenshot shows the Microsoft Excel interface with the 'Find and Replace' dialog box open. The dialog is configured to find the text '1983' and replace it with '1983'. The 'Look in' dropdown is set to 'Formulas'. The 'Find All' button is highlighted. The background spreadsheet shows a table with columns A through T and rows 1041 through 1061. The table contains data including dates, times, and tier numbers.

The screenshot shows the Microsoft Excel interface with the formula bar displaying the formula `=MODE(MATCH($H$2:$H$2336,$H$2:$H$2336,0))`. The spreadsheet has columns A through Q and rows 1 through 21. The columns are labeled: Customer ID, BMI, HBAIC, Heart Issues, Any Transplants, Cancer history, NumberOfMajorSurgeries, and smoker. The data in the 'smoker' column is being calculated using the formula.

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J2327 =MATCH(\$H\$2:\$H\$2336,\$H\$2:\$H\$2336,0)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
2316	Id2315	24.14	5.29	No	yes	No			1	No	180						
2317	Id2316	25.03	5.91	No	yes	No			1	No	180						
2318	Id2317	20.47	5.81	yes	No	No			1	No	180						
2319	Id2318	18.82	5.51	yes	No	No	No major surgery		No		180						
2320	Id2319	17.08	5.73	No	No	Yes			1	No	180						
2321	Id2320	19.21	5.53	yes	No	No	No major surgery		No		180						
2322	Id2321	17.86	5.43	No	No	Yes			1	No	180						
2323	Id2322	21.38	8.01	No	No	No	No major surgery		No		180						
2324	Id2323	23.35	5.94	No	No	No	No major surgery		No		180						
2325	Id2324	22.24	5.04	No	No	No	No major surgery		No		180						
2326	Id2325	24.76	4.54	yes	No	No	No major surgery		No		180						
2327	Id2326	20.1	5.6	yes	No	Yes			1	No	180						
2328	Id2327	21.77	10.67	No	No	No	No major surgery		No		180						
2329	Id2328	17.82	5.26	yes	No	No			1	No	180						
2330	Id2329	17.07	5.22	No	No	Yes			1	No	180						
2331	Id2330	22.24	4.29	yes	No	No	No major surgery		No		180						
2332	Id2331	22.34	5.57	No	No	No			1	No	180						
2333	Id2332	17.7	6.28	No	No	No			1	No	180						
2334	Id2333	16.47	6.35	No	No	Yes			1	No	180						
2335	Id2334	17.6	4.39	No	No	No			1	No	180						
2336	Id2335	17.58	4.51	No	No	No			1	No	180						

Customer Names Medical Examinations Hospitalisation Details

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H12 tier - 2

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	Customer	year	month	date	children	charge	Hospital tier	City tier	State											
2	Id2335	1992	Jul	9	0	563.84	tier - 2	tier - 3	R1013	1	9									
3	Id2334	1992	Nov	30	0	570.62	tier - 2	tier - 1	R1013	2										
4	Id2333	1993	Jun	30	0	600	tier - 2	tier - 1	R1013	2										
5	Id2332	1992	Sep	13	0	604.54	tier - 3	tier - 3	R1013	1										
6	Id2331	1998	Jul	27	0	637.26	tier - 3	tier - 3	R1013	1										
7	Id2330	2001	Nov	20	0	646.14	tier - 3	tier - 3	R1012	1										
8	Id2329	1993	Jun	1	0	650	tier - 3	tier - 3	R1013	1										
9	Id2328	1995	Jul	4	0	650	tier - 3	tier - 3	R1013	1										
10	Id2327	2002	Nov	29	0	668	tier - 3	tier - 2	R1012	9										
11	Id2326	1997	Nov	9	0	670	tier - 3	tier - 3	R1013	1										
12	Id2325	2001	Sep	12	0	687.54	tier - 3	tier - 2	R1013	9										
13	Id2324	1999	Dec	26	0	700	tier - 2	tier - 3	R1013	1										
14	Id2323	1999	Dec	14	0	722.99	tier - 3	tier - 1	R1013	2										
15	Id2322	2002	Sep	19	0	750	tier - 3	tier - 1	R1012	2										
16	Id2321	1993	Aug	9	0	760	tier - 3	tier - 1	R1013	2										
17	Id2320	1996	Oct	22	0	760	tier - 3	tier - 3	R1013	1										
18	Id2319	1993	Jun	28	0	770	tier - 3	tier - 3	R1013	1										
19	Id2318	1996	Sep	18	0	770.38	tier - 3	?	R1012	18										
20	Id2317	1995	Dec	7	0	773.54	tier - 3	tier - 2	R1013	9										
21	Id2316	2004	Oct	7	0	830.52	tier - 3	tier - 2	R1011	9										

Medical Examinations Hospitalisation Details Hospitalisation Details (2)

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Average: 9 Count: 4 Sum: 9

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Microsoft Excel interface showing a spreadsheet with columns: Customer, year, month, date, childn, charg, Hospital t, City ti, State. Row 1544 is highlighted. The State column contains 'Unknown'.

	Customer	year	month	date	childn	charg	Hospital t	City ti	State
1544	Id1793	1995	Dec	1	3	4827.9 tier - 1	tier - 2	Unknown	
2169	Id170	2000	Sep	5	1	37165.16 tier - 1	tier - 3	?	

Data Cleaning – Question 4

Question: If any 'State ID' values are missing, consider filling them with 'Unknown' or using another appropriate strategy.

Explanation: This means check State ID for missing values. If the correct State ID cannot be reliably determined, use Unknown or another justified method.

Technique used: Filter/check the State ID column and enter an appropriate replacement for missing values.

Formula used: No formula is specified or visibly required in the source for this question.

Microsoft Excel interface showing a spreadsheet with columns: Customer, year, month, date, childn, charg, Hospital t, City ti, State. Row J2 is highlighted. The formula bar shows `=MATCH($G$2:$G$2338,$G$2:$G$2338,0)`.

	Customer	year	month	date	childn	charg	Hospital t	City ti	State
2	Id2335	1992	Jul	9	0	563.84 tier - 2	tier - 3	R1013	1
3	Id2334	1992	Nov	30	0	570.62 tier - 2	tier - 1	R1013	1
4	Id2333	1993	Jun	30	0	600 tier - 2	tier - 1	R1013	1
5	Id2332	1992	Sep	13	0	604.54 tier - 3	tier - 3	R1013	4
6	Id2331	1998	Jul	27	0	637.26 tier - 3	tier - 3	R1013	4
7	Id2330	2001	Nov	20	0	646.14 tier - 3	tier - 3	R1012	4
8	Id2329	1993	Jun	1	0	650 tier - 3	tier - 3	R1013	4
9	Id2328	1995	Jul	4	0	650 tier - 3	tier - 3	R1013	4
10	Id2327	2002	Nov	29	0	668 tier - 3	tier - 2	R1012	4
11	Id2326	1997	Nov	9	0	670 tier - 3	tier - 3	R1013	4
12	Id2325	2001	Sep	12	0	687.54 tier - 3	tier - 2	R1013	4
13	Id2324	1999	Dec	26	0	700 ?	tier - 3	R1013	12
14	Id2323	1999	Dec	14	0	722.99 tier - 3	tier - 1	R1013	4
15	Id2322	2002	Sep	19	0	750 tier - 3	tier - 1	R1012	4
16	Id2321	1993	Aug	9	0	760 tier - 3	tier - 1	R1013	4
17	Id2320	1996	Oct	22	0	760 tier - 3	tier - 3	R1013	4
18	Id2319	1993	Jun	28	0	770 tier - 3	tier - 3	R1013	4
19	Id2318	1996	Sep	18	0	770.38 tier - 3	?	R1012	4
20	Id2317	1995	Dec	7	0	773.54 tier - 3	tier - 2	R1013	4
21	Id2316	2004	Oct	7	0	830.52 tier - 3	tier - 2	R1011	4

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Calibri 11 A⁺ A⁻ B I U % 0.00 0.00

General Conditional Formatting Format as Table Cell Styles Insert Delete Format Sort & Filter Find & Select Add-ins

K2 =MODE(MATCH(\$G\$2:\$G\$2338,\$G\$2:\$G\$2338,0))

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	Customer	year	month	date	children	charge	Hospital t	City ti	State											
2	Id2335	1992	Jul	9	0	563.84	tier - 2	tier - 3	R1013	1										
3	Id2334	1992	Nov	30	0	570.62	tier - 2	tier - 1	R1013	1										
4	Id2333	1993	Jun	30	0	600	tier - 2	tier - 1	R1013	1										
5	Id2332	1992	Sep	13	0	604.54	tier - 3	tier - 3	R1013	4										
6	Id2331	1998	Jul	27	0	637.26	tier - 3	tier - 3	R1013	4										
7	Id2330	2001	Nov	20	0	646.14	tier - 3	tier - 3	R1012	4										
8	Id2329	1993	Jun	1	0	650	tier - 3	tier - 3	R1013	4										
9	Id2328	1995	Jul	4	0	650	tier - 3	tier - 3	R1013	4										
10	Id2327	2002	Nov	29	0	668	tier - 3	tier - 2	R1012	4										
11	Id2326	1997	Nov	9	0	670	tier - 3	tier - 3	R1013	4										
12	Id2325	2001	Sep	12	0	687.54	tier - 3	tier - 2	R1013	4										
13	Id2324	1999	Dec	26	0	700	?	tier - 3	R1013	12										
14	Id2323	1999	Dec	14	0	722.99	tier - 3	tier - 1	R1013	4										
15	Id2322	2002	Sep	19	0	750	tier - 3	tier - 1	R1012	4										
16	Id2321	1993	Aug	9	0	760	tier - 3	tier - 1	R1013	4										
17	Id2320	1996	Oct	22	0	760	tier - 3	tier - 3	R1013	4										
18	Id2319	1993	Jun	28	0	770	tier - 3	tier - 3	R1013	4										
19	Id2318	1996	Sep	18	0	770.38	tier - 3	?	R1012	4										
20	Id2317	1995	Dec	7	0	773.54	tier - 3	tier - 2	R1013	4										
21	Id2316	2004	Oct	7	0	830.52	tier - 3	tier - 2	R1011	4										

Ready Accessibility: Good to go

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Data Transformation

Data Transformation – Question 1

Question: Split the ‘names’ column in the “Customer Names” table into 3 meaningful columns: ‘Title’, ‘First Name’, and ‘Last Name’.

Explanation: This means separate the single combined name field into three usable fields.

Technique used: Excel Data → Text to Columns. The screenshots show the Convert Text to Columns Wizard and the resulting separated columns.

Formula used: No formula is used in the screenshot; the technique is Text to Columns.

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Get & Transform Data Queries & Connections Data Types Sort & Filter Data Tools Forecast Analysis

Convert Text to Columns Wizard - Step 1 of 3

This screen lets you select each column and set the Data Format.

Column data format

☒ General
☐ Text
☐ Date: DMY
☐ Do not import column (skip)

'General' converts numeric values to numbers, date values to dates, and all remaining values to text.

Advanced...

Destination: =\$C\$1

Data preview

General	General
name	
Hawks	Ms. Kelly
Lehner	Mr. Matthew D
Lu	Mr. Phil
Osborne	Ms. Kelsey

Cancel < Back Next > Finish

Customer ID name

Customer ID	name
Id1	Hawks, Ms. Kelly
Id2	Lehner, Mr. Matthew D
Id3	Lu, Mr. Phil
Id4	Osborne, Ms. Kelsey
Id5	Kadala, Ms. Kristyn
Id6	Baker, Mr. Russell B.
Id7	Macpherson, Mr. Scott
Id8	Hallman, Mr. Stephen
Id9	Moran, Mr. Patrick R.
Id10	Benner, Ms. Brooke N.
Id11	Fierro Vargas, Ms. Paola Andrea
Id12	Franz, Mr. David
Id13	Foster, Mr. Wade
Id14	Tenorio, Mr. Franklin
Id15	Rios, Ms. Leilani M.
Id16	Viau-Dupuis, Mr. Philippe
Id17	Cronin, Ms. Jennifer A.
Id18	Noordstar, Ms. Christina M.
Id19	Boudalia, Mr. Said Sr.

Customer Names Medical Examinations Hospitalisation Details

Enter Accessibility: Good to go

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Get & Transform Data Queries & Connections Data Types Sort & Filter Data Tools Forecast Analysis

Convert Text to Columns Wizard - Step 1 of 3

This screen lets you select each column and set the Data Format.

Column data format

☒ General
☐ Text
☐ Date: DMY
☐ Do not import column (skip)

'General' converts numeric values to numbers, date values to dates, and all remaining values to text.

Advanced...

Destination: =\$C\$1

Data preview

General	General
name	
Hawks	Ms. Kelly
Lehner	Mr. Matthew D
Lu	Mr. Phil
Osborne	Ms. Kelsey

Cancel < Back Next > Finish

Customer ID name

Customer ID	name	name
Id1	Hawks, Ms. Kelly	Hawks Ms. Kelly
Id2	Lehner, Mr. Matthew D	Lehner Mr. Matthew D
Id3	Lu, Mr. Phil	Lu Mr. Phil
Id4	Osborne, Ms. Kelsey	Osborne Ms. Kelsey
Id5	Kadala, Ms. Kristyn	Kadala Ms. Kristyn
Id6	Baker, Mr. Russell B.	Baker Mr. Russell B.
Id7	Macpherson, Mr. Scott	Macphers Mr. Scott
Id8	Hallman, Mr. Stephen	Hallman Mr. Stephen
Id9	Moran, Mr. Patrick R.	Moran Mr. Patrick R.
Id10	Benner, Ms. Brooke N.	Benner Ms. Brooke N.
Id11	Fierro Vargas, Ms. Paola Andrea	Fierro Var Ms. Paola Andrea
Id12	Franz, Mr. David	Franz Mr. David
Id13	Foster, Mr. Wade	Foster Mr. Wade
Id14	Tenorio, Mr. Franklin	Tenorio Mr. Franklin
Id15	Rios, Ms. Leilani M.	Rios Ms. Leilani M.
Id16	Viau-Dupuis, Mr. Philippe	Viau-Dup Mr. Philippe
Id17	Cronin, Ms. Jennifer A.	Cronin Ms. Jennifer A.
Id18	Noordstar, Ms. Christina M.	Noordstar Ms. Christina M.
Id19	Boudalia, Mr. Said Sr.	Boudalia Mr. Said Sr.

Customer Names Medical Examinations Hospitalisation Details

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Count: 2336

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Search

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Convert Text to Columns Wizard - Step 2 of 3

This screen lets you set field widths (column breaks). Lines with arrows signify a column break.

To CREATE a break line, click at the desired position.
To DELETE a break line, double click on the line.
To MOVE a break line, click and drag it.

Data preview

Ms. Kelly
Mr. Matthew D
Mr. Phil
Ms. Kelsey

Buttons: Cancel, < Back, Next >, Finish

Customer ID	name	First Name	Last Name
Id1	Hawks, Ms. Kelly	Kelly	Hawks
Id2	Lehner, Mr. Matthew D	Matthew D	Lehner
Id3	Lu, Mr. Phil	Phil	Lu
Id4	Osborne, Ms. Kelsey	Kelsey	Osborne
Id5	Kadala, Ms. Kristyn	Kristyn	Kadala
Id6	Baker, Mr. Russell B.	Russell B.	Baker
Id7	Macpherson, Mr. Scott	Scott	Macpherson
Id8	Hallman, Mr. Stephen	Stephen	Hallman
Id9	Moran, Mr. Patrick R.	Patrick R.	Moran
Id10	Benner, Ms. Brooke N.	Brooke N.	Benner
Id11	Fierro Vargas, Ms. Paola Andrea	Paola Andrea	Fierro Vargas
Id12	Franz, Mr. David	David	Franz
Id13	Foster, Mr. Wade	Wade	Foster
Id14	Tenorio, Mr. Franklin	Franklin	Tenorio
Id15	Rios, Ms. Lellani M.	Lellani M.	Rios
Id16	Viau-Dupuis, Mr. Philippe	Philippe	Viau-Dupuis
Id17	Cronin, Ms. Jennifer A.	Jennifer A.	Cronin
Id18	Noordstar, Ms. Christina M.	Christina M.	Noordstar
Id19	Boudalia, Mr. Said Sr.	Said Sr.	Boudalia

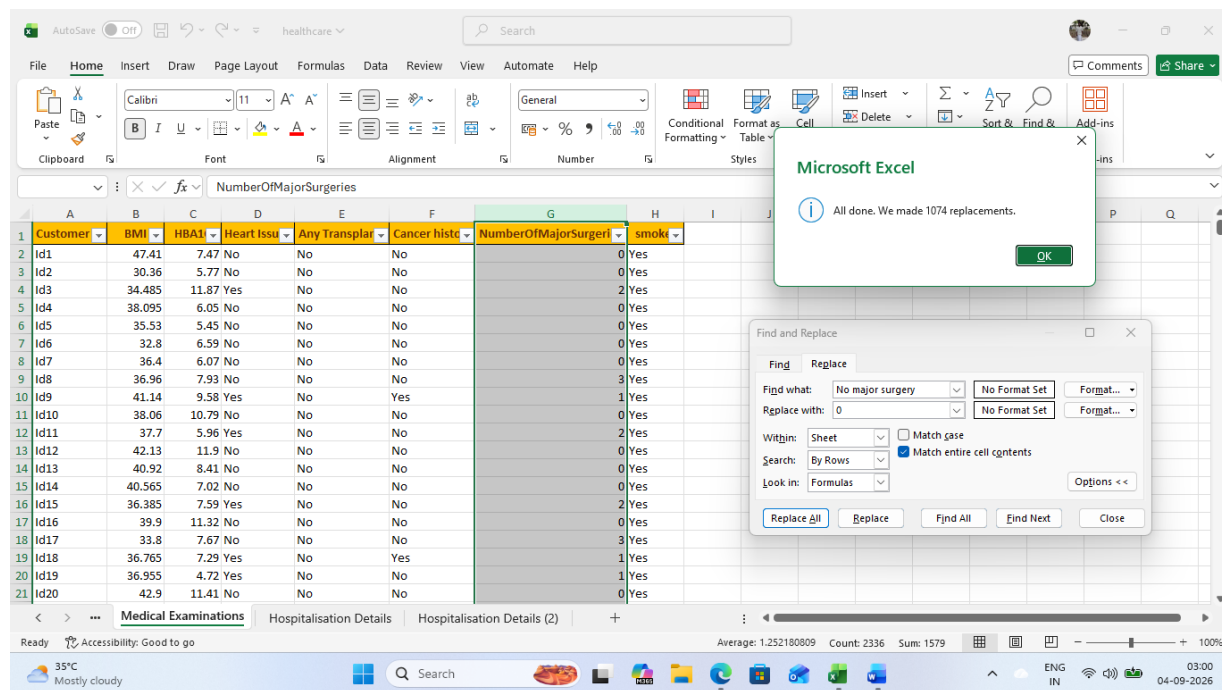
Data Transformation – Question 2

Question: Convert the "NumberOfMajorSurgeries" column in the "Medical Examinations" table to numerical data by replacing non-numeric characters with meaningful numerical values.

Explanation: This means change text such as 'No major surgery' into the appropriate numeric value so the entire field can be calculated.

Technique used: Find & Replace is used in the screenshot to replace 'No major surgery' with 0.

Formula used: Exact replacement shown: Find 'No major surgery' → Replace with '0'.



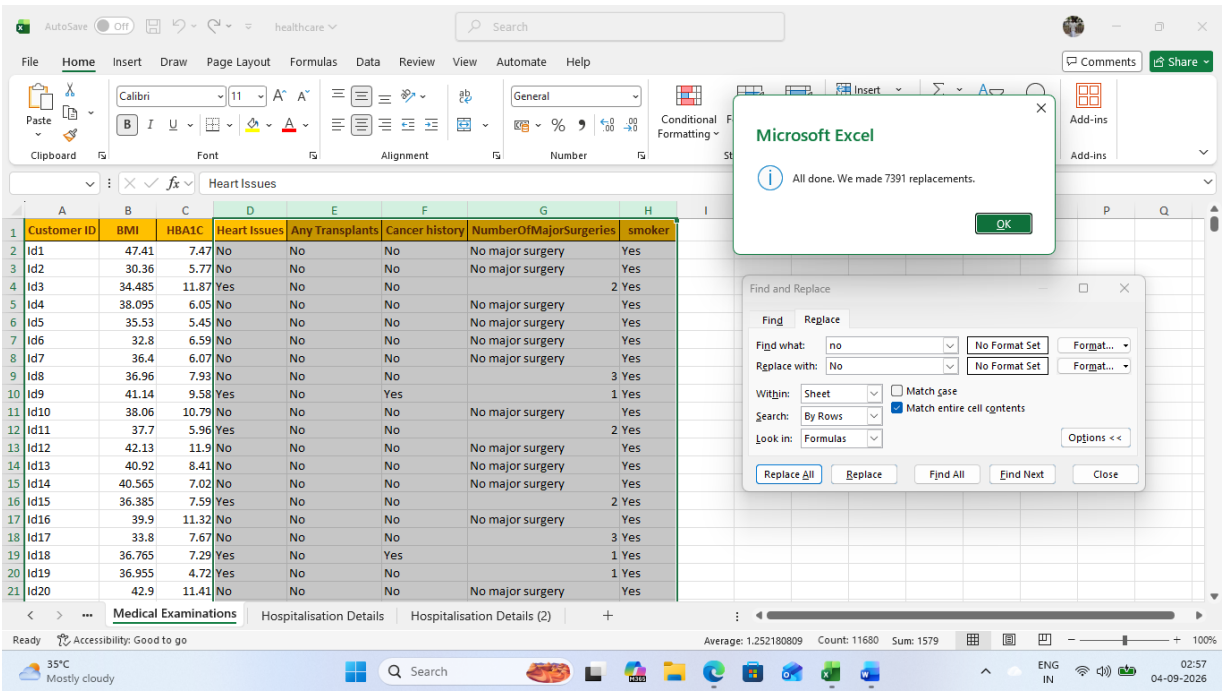
Data Transformation – Question 3

Question: Check for inconsistencies in the 'Heart Issues' and 'Smoker' columns and propose corrective actions if necessary.

Explanation: This means inspect the category values and make equivalent responses consistent, for example standardizing No/yes/Yes into one consistent representation.

Technique used: Find & Replace is shown in the screenshot. The source screenshot shows 'no' being replaced with 'No'.

Formula used: Exact replacement shown: Find 'no' → Replace with 'No'.



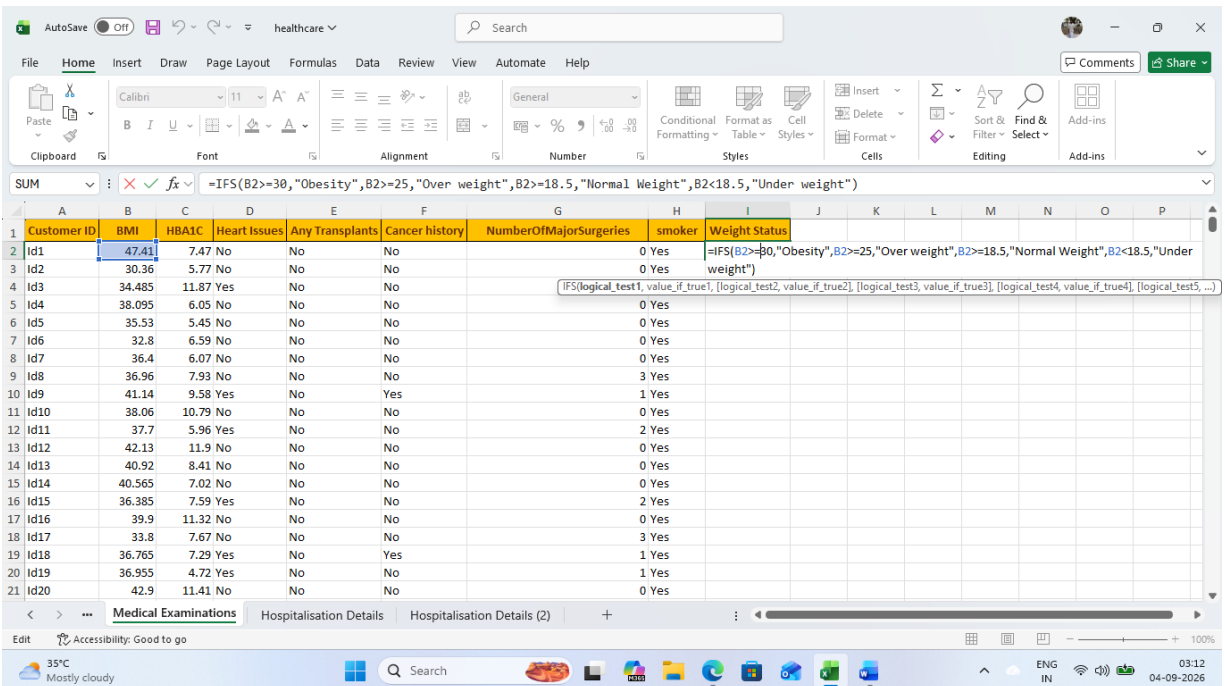
Data Transformation – Question 4

Question: Create a new column named “Weight Status” that categorizes BMI into different categories as below:

Explanation: This means convert each BMI value into the Weight Status category specified by the assignment.

Technique used: IFS function is used in the source screenshot to classify BMI into four categories.

Formula used: Exact formula visible: `=IFS(B2>=30,"Obesity",B2>=25,"Over weight",B2>=18.5,"Normal Weight",B2<18.5,"Under weight")`



BMI categories from the original question

BMI	Weight Status
Below 18.5	Underweight
18.5 – 24.9	Normal Weight
25.0 – 29.9	Overweight
30.0 and Above	Obesity

Data Transformation – Question 5

Question: Merge the ‘year’, ‘month’, and ‘date’ columns in the “Hospitalization Details” table into one column named ‘Date of Birth’ and format it in the ‘DD-MMM-YYYY’ custom format.

Explanation: This means create one real date from Year, Month and Date, call the new field Date of Birth, and display it in the required format.

Technique used: CONCAT is used to combine the date parts, DATEVALUE converts the text to an Excel date, and the result is displayed as DD-MMM-YYYY.

Formula used: Exact formula visible in the source screenshot: =DATEVALUE(TEXT(CONCAT(D11,C11,B11),"DD-MM-YYYY"))

Customer ID	year	month	date	children	charges	Hospital tier	City tier	State ID	Date of Birth	Age
Id2335	1992	Jul	9	0	\$ 563.84	tier - 2	tier - 3	R1013	09-07-1992	31
Id2334	1992	Nov	30	0	\$ 570.62	tier - 2	tier - 1	R1013	30-11-1992	31
Id2333	1993	Jun	30	0	\$ 600.00	tier - 2	tier - 1	R1013	30-06-1993	30
Id2332	1992	Sep	13	0	\$ 604.54	tier - 3	tier - 3	R1013	13-09-1992	31
Id2331	1998	Jul	27	0	\$ 637.26	tier - 3	tier - 3	R1013	27-07-1998	25
Id2330	2001	Nov	20	0	\$ 646.14	tier - 3	tier - 3	R1012	20-11-2001	22
Id2329	1993	Jun	1	0	\$ 650.00	tier - 3	tier - 3	R1013	01-06-1993	30
Id2328	1995	Jul	4	0	\$ 650.00	tier - 3	tier - 3	R1013	04-07-1995	28
Id2327	2002	Nov	29	0	\$ 668.00	tier - 3	tier - 2	R1012	29-11-2002	21
Id2326	1997	Nov	9	0	\$ 670.00	tier - 3	tier - 3	R1013	09-11-1997	26
Id2325	2001	Sep	12	0	\$ 687.54	tier - 3	tier - 2	R1013	12-09-2001	22
Id2324	1999	Dec	26	0	\$ 700.00	tier - 2	tier - 3	R1013	26-12-1999	24
Id2323	1999	Dec	14	0	\$ 722.99	tier - 3	tier - 1	R1013	14-12-1999	24
Id2322	2002	Sep	19	0	\$ 750.00	tier - 3	tier - 1	R1012	19-09-2002	21
Id2321	1993	Aug	9	0	\$ 760.00	tier - 3	tier - 1	R1013	09-08-1993	30
Id2320	1996	Oct	22	0	\$ 760.00	tier - 3	tier - 3	R1013	22-10-1996	27
Id2319	1993	Jun	28	0	\$ 770.00	tier - 3	tier - 3	R1013	28-06-1993	30
Id2318	1996	Sep	18	0	\$ 770.38	tier - 3	tier - 2	R1012	18-09-1996	27
Id2317	1995	Dec	7	0	\$ 773.54	tier - 3	tier - 2	R1013	07-12-1995	28
Id2316	2004	Oct	7	0	\$ 830.52	tier - 3	tier - 2	R1011	07-10-2004	19
Id2315	2000	Nov	18	0	\$ 865.41	tier - 3	tier - 1	R1013	18-11-2000	23

Data Transformation – Question 7

Question: Calculate the ‘Age’ of each customer based on their ‘Date of Birth’ and the date of collection of the dataset, which is 8th June 2023.

Explanation: This means calculate the customer's age using the Date of Birth and the fixed collection date 8 June 2023.

Technique used: YEAR and DATE are used in the source screenshot.

Formula used: Exact formula visible: =YEAR(DATE(2023,6,8))-YEAR(J2)

The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
1	Customer ID	year	month	date	children	charges	Hospital tier	City tier	State ID	Date of Birth	Age									
2	Id2335	1992	Jul	9	0	563.84	tier - 2	tier - 3	R1013	09-07-1992	=YEAR(DATE(2023,6,8))-YEAR(J2)									
3	Id2334	1992	Nov	30	0	570.62	tier - 2	tier - 1	R1013	30-11-1992										
4	Id2333	1993	Jun	30	0	600	tier - 2	tier - 1	R1013	30-06-1993										
5	Id2332	1992	Sep	13	0	604.54	tier - 3	tier - 3	R1013	13-09-1992										
6	Id2331	1998	Jul	27	0	637.26	tier - 3	tier - 3	R1013	27-07-1998										
7	Id2330	2001	Nov	20	0	646.14	tier - 3	tier - 3	R1012	20-11-2001										
8	Id2329	1993	Jun	1	0	650	tier - 3	tier - 3	R1013	01-06-1993										
9	Id2328	1995	Jul	4	0	650	tier - 3	tier - 3	R1013	04-07-1995										
10	Id2327	2002	Nov	29	0	668	tier - 3	tier - 2	R1012	29-11-2002										
11	Id2326	1997	Nov	9	0	670	tier - 3	tier - 3	R1013	09-11-1997										
12	Id2325	2001	Sep	12	0	687.54	tier - 3	tier - 2	R1013	12-09-2001										
13	Id2324	1999	Dec	26	0	700	tier - 2	tier - 3	R1013	26-12-1999										
14	Id2323	1999	Dec	14	0	722.99	tier - 3	tier - 1	R1013	14-12-1999										
15	Id2322	2002	Sep	19	0	750	tier - 3	tier - 1	R1012	19-09-2002										
16	Id2321	1993	Aug	9	0	760	tier - 3	tier - 1	R1013	09-08-1993										
17	Id2320	1996	Oct	22	0	760	tier - 3	tier - 3	R1013	22-10-1996										
18	Id2319	1993	Jun	28	0	770	tier - 3	tier - 3	R1013	28-06-1993										
19	Id2318	1996	Sep	18	0	770.38	tier - 3	tier - 2	R1012	18-09-1996										
20	Id2317	1995	Dec	7	0	773.54	tier - 3	tier - 2	R1013	07-12-1995										
21	Id2316	2004	Oct	7	0	830.52	tier - 3	tier - 2	R1011	07-10-2004										

Data Transformation – Question 8

Question: Format the 'charges' column as currency (\$).

Explanation: This means change the display of Charges into dollar currency while keeping the values numeric.

Technique used: Excel Accounting/Currency number formatting. The source screenshot shows the currency selection menu.

Formula used: No calculation formula; this is number formatting.

Customer ID	year	month	date	children	charges	Hospital tier	City tier	State
Id2335	1992	Jul	9	0	563.84	tier - 2	tier - 3	R1013
Id2334	1992	Nov	30	0	570.62	tier - 2	tier - 1	R1013
Id2333	1993	Jun	30	0	600	tier - 2	tier - 1	R1013
Id2332	1992	Sep	13	0	604.54	tier - 3	tier - 3	R1013
Id2331	1998	Jul	27	0	637.26	tier - 3	tier - 3	R1013
Id2330	2001	Nov	20	0	646.14	tier - 3	tier - 3	R1012
Id2329	1993	Jun	1	0	650	tier - 3	tier - 3	R1013
Id2328	1995	Jul	4	0	650	tier - 3	tier - 3	R1013
Id2327	2002	Nov	29	0	668	tier - 3	tier - 2	R1012
Id2326	1997	Nov	9	0	670	tier - 3	tier - 3	R1013
Id2325	2001	Sep	12	0	687.54	tier - 3	tier - 2	R1013
Id2324	1999	Dec	26	0	700	tier - 2	tier - 3	R1013
Id2323	1999	Dec	14	0	722.99	tier - 3	tier - 1	R1013
Id2322	2002	Sep	19	0	750	tier - 3	tier - 1	R1012
Id2321	1993	Aug	9	0	760	tier - 3	tier - 1	R1013
Id2320	1996	Oct	22	0	760	tier - 3	tier - 3	R1013
Id2319	1993	Jun	28	0	770	tier - 3	tier - 3	R1013
Id2318	1996	Sep	18	0	770.38	tier - 3	tier - 2	R1012
Id2317	1995	Dec	7	0	773.54	tier - 3	tier - 2	R1013
Id2316	2004	Oct	7	0	830.52	tier - 3	tier - 2	R1011

AI-Driven Data Exploration, Analysis & Visualization

Activity 1: Data Preparation (should be done manually)

Question: Create a new sheet named “Healthcare”, and combine all three tables into one, using Customer ID as the common column, utilizing XLOOKUP/Data Model manager.

Explanation: This means create one combined Healthcare sheet and use Customer ID to bring information from the separate tables into the same record.

Technique used: XLOOKUP / Data Model manager. The source screenshot explicitly shows XLOOKUP being used.

Formula used: Exact XLOOKUP formula visible: =XLOOKUP([@[Customer ID]],Table3[Customer ID],Table3[BMI])

DD_healthcare - Saved

Search

File Home Insert Draw Page Layout Formulas Data Review View Automate Help Data Streamer Power Pivot Table Design

Clipboard Font Alignment Number Styles Cells Editing Add-ins

C2: =XLOOKUP([@Customer ID],Table3[Customer ID],Table3[BMI])

	A	B	C	D	E	F	G	H	I	J	K	L
	Customer ID	First Name	BMI	HBA1	Heart Issue	Any Transplant	Cancer Histor	Number of Major Surgeries	Smoke	Weight Statu	Date of Birt	Charge
1	Id1	Kelly	47.41	7.47	No	No	No	0	Yes	Obesity	12-10-1968	63,770.43
2	Id2	Matthew D	30.36	5.77	No	No	No	0	Yes	Obesity	08-06-1977	62,592.87
3	Id3	Phil	34.485	11.87	Yes	No	No	2	Yes	Obesity	11-09-1970	60,021.40
4	Id4	Kelsey	38.095	6.05	No	No	No	0	Yes	Obesity	06-06-1991	58,571.07
5	Id5	Kristyn	35.53	5.45	No	No	No	0	Yes	Obesity	19-06-1989	55,135.40
6	Id6	Russell B	32.8	6.59	No	No	No	0	Yes	Obesity	04-08-1962	52,590.83
7	Id7	Scott	36.4	6.07	No	No	No	0	Yes	Obesity	27-10-1994	51,194.56
8	Id8	Stephen	36.96	7.93	No	No	No	3	Yes	Obesity	27-06-1958	49,577.66
9	Id9	Patrick R	41.14	9.58	Yes	No	Yes	1	Yes	Obesity	04-09-1963	48,970.25
10	Id10	Brooke N	38.06	10.79	No	No	No	0	Yes	Obesity	29-12-1978	48,885.14
11	Id11	Paola Andrea	37.7	5.96	Yes	No	No	2	Yes	Obesity	22-07-1959	48,824.45
12	Id12	David	42.13	11.9	No	No	No	0	Yes	Obesity	27-10-1965	48,675.52
13	Id13	Wade	40.92	8.41	No	No	No	0	Yes	Obesity	11-10-1962	48,673.56
14	Id14	Franklin	40.565	7.02	No	No	No	0	Yes	Obesity	01-12-1968	48,549.18
15	Id15	Leilani M	36.385	7.59	Yes	No	No	2	Yes	Obesity	21-12-1961	48,517.56
16	Id16	Phillippe	39.9	11.32	No	No	No	0	Yes	Obesity	27-08-1962	48,173.36
17	Id17	Jennifer A	33.8	7.67	No	No	No	3	Yes	Obesity	16-11-1958	47,928.03

Customer Names Medical Examinations Hospitalisation Details Healthcare

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Activity 1: Data Preparation (should be done manually) – Required columns

Question: Retain the following necessary columns: Customer ID, First Name, BMI, HBA1C, Heart Issues, Any Transplants, Cancer History, Number of Major Surgeries, Smoker, Weight Status, Diabetes Status, Date of Birth, Charges, Hospital Tier, City Tier, State ID, and Age.

Explanation: This means keep only the listed analysis fields in the combined Healthcare dataset.

Technique used: Column selection/retention in the combined Healthcare sheet.

Formula used: No formula is specified by this question itself.

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Search

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Clipboard Font Alignment Number Styles Cells Editing Add-ins

C2: =XLOOKUP([@Customer ID],Table3[Customer ID],Table3[BMI])

	A	B	C	D	E	F	G	H	I	J	K	L
	Customer ID	First Name	BMI	HBA1	Heart Issue	Any Transplant	Cancer Histor	Number of Major Surgeries	Smoke	Weight Statu	Date of Birt	Charge
1	Id1	Kelly	47.41	7.47	No	No	No	0	Yes	Obesity	12-10-1968	63,770.43
2	Id2	Matthew D	30.36	5.77	No	No	No	0	Yes	Obesity	08-06-1977	62,592.87
3	Id3	Phil	34.485	11.87	Yes	No	No	2	Yes	Obesity	11-09-1970	60,021.40
4	Id4	Kelsey	38.095	6.05	No	No	No	0	Yes	Obesity	06-06-1991	58,571.07
5	Id5	Kristyn	35.53	5.45	No	No	No	0	Yes	Obesity	19-06-1989	55,135.40
6	Id6	Russell B	32.8	6.59	No	No	No	0	Yes	Obesity	04-08-1962	52,590.83
7	Id7	Scott	36.4	6.07	No	No	No	0	Yes	Obesity	27-10-1994	51,194.56
8	Id8	Stephen	36.96	7.93	No	No	No	3	Yes	Obesity	27-06-1958	49,577.66
9	Id9	Patrick R	41.14	9.58	Yes	No	Yes	1	Yes	Obesity	04-09-1963	48,970.25
10	Id10	Brooke N	38.06	10.79	No	No	No	0	Yes	Obesity	29-12-1978	48,885.14
11	Id11	Paola Andrea	37.7	5.96	Yes	No	No	2	Yes	Obesity	22-07-1959	48,824.45
12	Id12	David	42.13	11.9	No	No	No	0	Yes	Obesity	27-10-1965	48,675.52
13	Id13	Wade	40.92	8.41	No	No	No	0	Yes	Obesity	11-10-1962	48,673.56
14	Id14	Franklin	40.565	7.02	No	No	No	0	Yes	Obesity	01-12-1968	48,549.18
15	Id15	Leilani M	36.385	7.59	Yes	No	No	2	Yes	Obesity	21-12-1961	48,517.56
16	Id16	Phillippe	39.9	11.32	No	No	No	0	Yes	Obesity	27-08-1962	48,173.36
17	Id17	Jennifer A	33.8	7.67	No	No	No	3	Yes	Obesity	16-11-1958	47,928.03

Customer Names Medical Examinations Hospitalisation Details Healthcare

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	A	B	C	D	E	F	G	H	I	J	K	L	M	N
2319	Id2318	Candice M	18.82	5.51	Yes	No	No		0	No	Normal Weight	18-09-1996	770.38	tier - 3
2320	Id2319	Theodore A	17.08	5.73	No	No	Yes		1	No	Under weight	28-06-1993	770.00	tier - 3
2321	Id2320	Lindy	19.21	5.53	Yes	No	No		0	No	Normal Weight	22-10-1996	760.00	tier - 3
2322	Id2321	Thomas	17.86	5.43	No	No	Yes		1	No	Under weight	09-08-1993	760.00	tier - 1
2323	Id2322	Holly	21.38	8.01	No	No	No		0	No	Normal Weight	19-09-2002	750.00	tier - 1
2324	Id2323	Matthew	23.35	5.94	No	No	No		0	No	Normal Weight	14-12-1999	722.99	tier - 1
2325	Id2324	Meghan K	22.24	5.04	No	No	No		0	No	Normal Weight	26-12-1999	700.00	tier - 2
2326	Id2325	Fred	24.76	4.54	Yes	No	No		0	No	Normal Weight	12-09-2001	687.54	tier - 2
2327	Id2326	Sebastian	20.1	5.6	Yes	No	Yes		1	No	Normal Weight	09-11-1997	670.00	tier - 3
2328	Id2327	Laura	21.77	10.67	No	No	No		0	No	Normal Weight	29-11-2002	668.00	tier - 3
2329	Id2328	Nicole	17.82	5.26	Yes	No	No		1	No	Under weight	04-07-1995	650.00	tier - 3
2330	Id2329	Susan E	17.07	5.22	No	No	Yes		1	No	Under weight	01-06-1993	650.00	tier - 3
2331	Id2330	Katy	22.24	4.29	Yes	No	No		0	No	Normal Weight	20-11-2001	646.14	tier - 3
2332	Id2331	Jordan	22.34	5.57	No	No	No		1	No	Normal Weight	27-07-1998	637.26	tier - 3
2333	Id2332	Juan D Sr	17.7	6.28	No	No	No		1	No	Under weight	13-09-1992	604.54	tier - 3
2334	Id2333	Julie	16.47	6.35	No	No	Yes		1	No	Under weight	30-06-1993	600.00	tier - 1
2335	Id2334	Evan P	17.6	4.39	No	No	No		1	No	Under weight	30-11-1992	570.62	tier - 1
2336	Id2335	Aaron K	17.58	4.51	No	No	No		1	No	Under weight	09-07-1992	563.84	tier - 2
2337	2335											13529.918		2335

Activity 2: Manual Dashboard Build (should be done manually)

Question: Create a pivot table manually (Insert → PivotTable) using the combined healthcare dataset.

Explanation: This means create a PivotTable from the Healthcare dataset so the required categories and totals can be summarized.

Technique used: Excel Insert → PivotTable, followed by PivotTable Fields.

Formula used: No formula; PivotTable technique.

AutoSave

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View

Search

FileHomeInsertDrawPage LayoutFormulasDataReviewViewAutomateHelpData StreamerPower PivotTable Design

PivotTableRecommended PivotTables

Tables

FormstIllustrations

Controls

Recommended Charts

Charts

MapsPivotChart

LineColumnWin/Loss

Sparklines

SlicerTimeline

Filters

LinkLinks

CommentComments

Text

Symbols

CommentsShare

A2

=XLOOKUP([@Customer ID],Table2[**#A11**],[Customer ID],Table2[**#A11**],[First Name])

	Customer ID	First Name	BMI	HBA1C	Heart Issues	Any Transplants	Cancer History	Number of Major Surgeries	Smoker	Weight Status	Date of Birth	Charges	
2	Id1	Kelly	47.41	7.47	No	No	No				12-10-1968	63,770.43	
3	Id2	Matthew D	30.36	5.77	No	No	No				08-06-1977	62,592.87	
4	Id3	Phil	34.485	11.87	Yes	No	No				11-09-1970	60,021.40	
5	Id4	Kelsey	38.095	6.05	No	No	No				06-06-1991	58,571.07	
6	Id5	Kristyn	35.53	5.45	No	No	No				19-06-1989	55,135.40	
7	Id6	Russell B	32.8	6.59	No	No	No				04-08-1962	52,590.83	
8	Id7	Scott	36.4	6.07	No	No	No				27-10-1994	51,194.56	
9	Id8	Stephen	36.96	7.93	No	No	No				27-06-1958	49,577.66	
10	Id9	Patrick R	41.14	9.58	Yes	No	Yes				04-09-1963	48,970.25	
11	Id10	Brooke N	38.06	10.79	No	No	No				29-12-1978	48,885.14	
12	Id11	Paola Andrea	37.7	5.96	Yes	No	No				22-07-1959	48,824.45	
13	Id12	David	42.13	11.9	No	No	No				27-10-1965	48,675.52	
14	Id13	Wade	40.92	8.41	No	No	No				11-10-1962	48,673.56	
15	Id14	Franklin	40.565	7.02	No	No	No				01-12-1968	48,549.18	
16	Id15	Leilani M	36.385	7.59	Yes	No	No		2	Yes	Obesity	21-12-1961	48,517.56
17	Id16	Phillippe	39.9	11.32	No	No	No		0	Yes	Obesity	27-08-1962	48,173.36
18	Id17	Jennifer A	33.8	7.67	No	No	No		3	Yes	Obesity	16-11-1958	47,928.03
19	Id18	Christina M	36.765	7.29	Yes	No	Yes		1	Yes	Obesity	05-08-1963	47,896.79

PivotTable from table or range

Select a table or range

Table/Range: **Table2**

Choose where you want the PivotTable to be placed

☒ New Worksheet

☐ Existing Worksheet

Locations

Choose whether you want to analyze multiple tables

☐ Add this data to the Data Model

OKCancel

Medical ExaminationsHospitalisation DetailsHealthcare

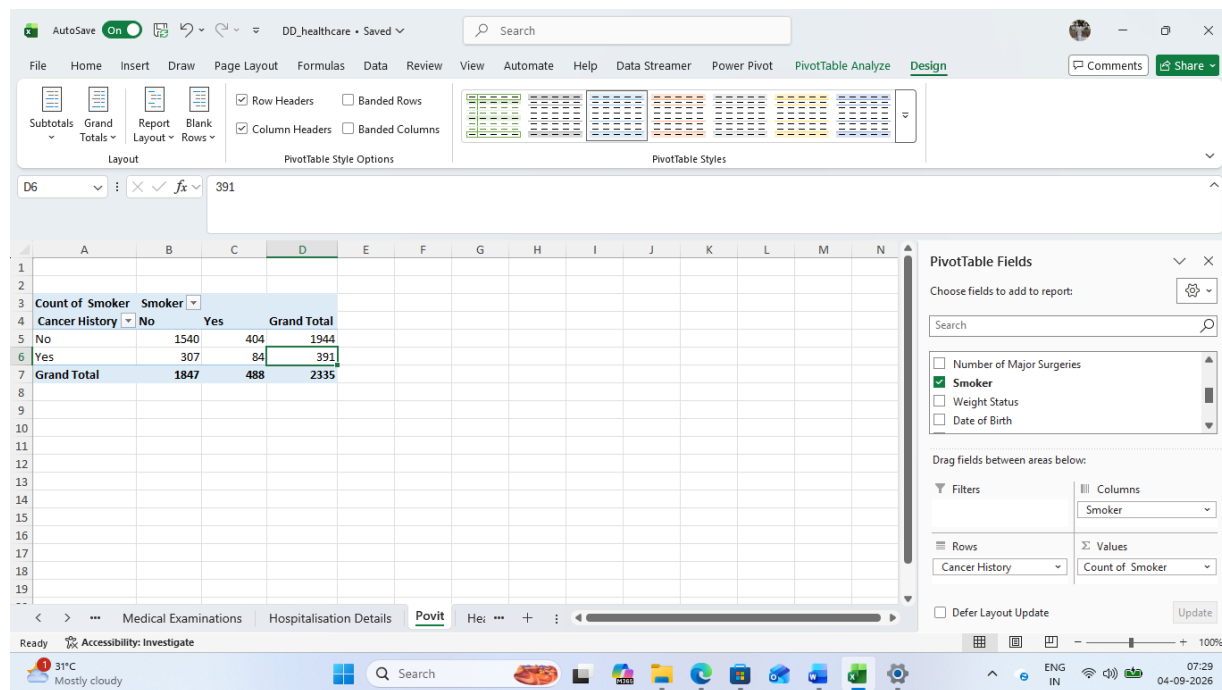
PointAccessibility: Investigate

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Activity 2: Manual Dashboard Build (should be done manually) – Charts

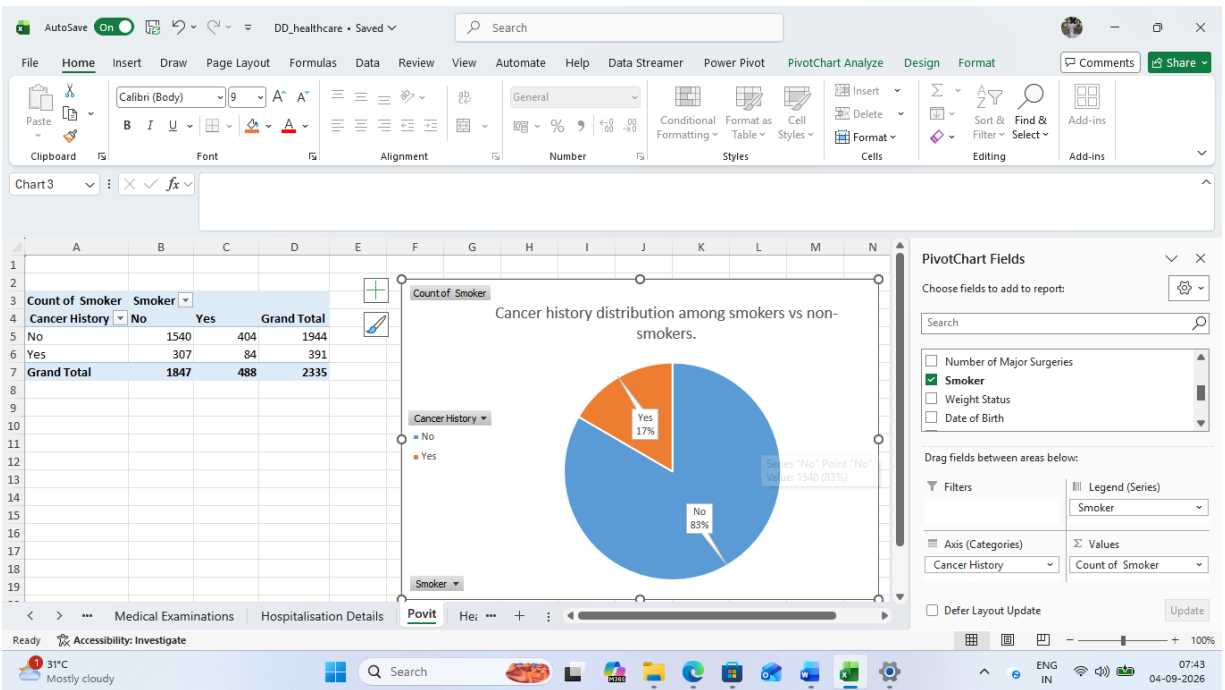
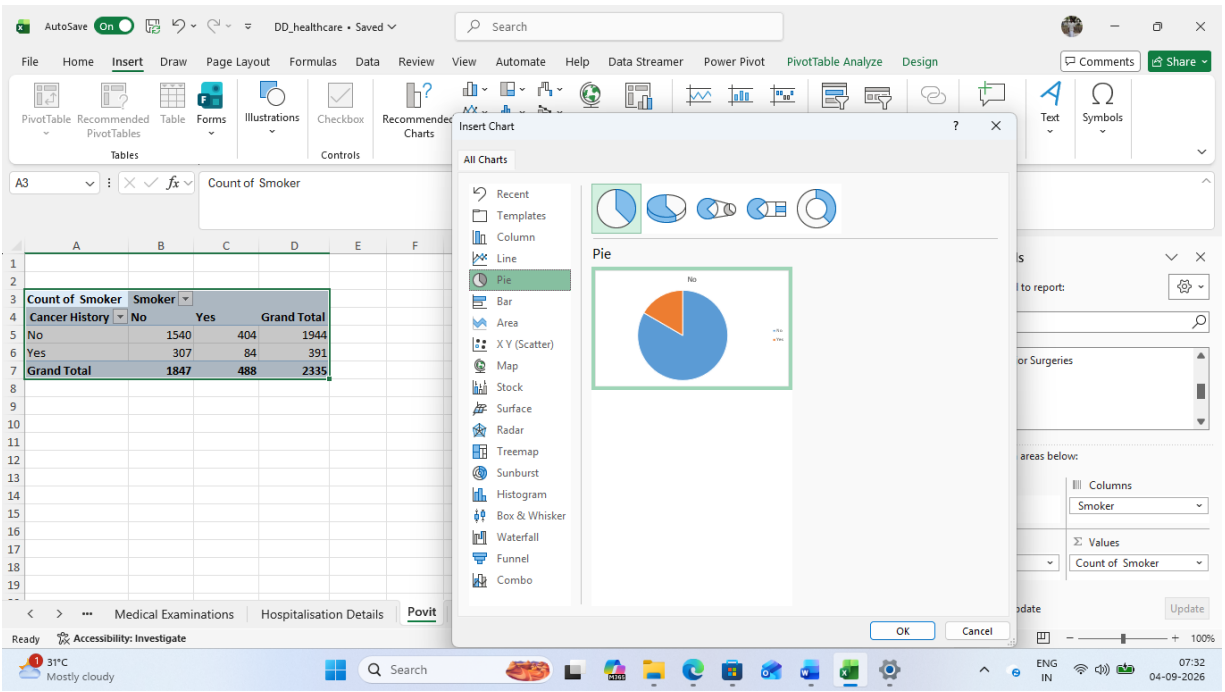
Question: Add charts:

- Pie/Donut Chart: Cancer history distribution among smokers vs non-smokers.
- Column Chart: Charges by Weight Status.

Explanation: This means turn the PivotTable summaries into the two specified visualizations.

Technique used: Insert → PivotChart/Chart. The source screenshots show the Pie chart selection and resulting chart.

Formula used: No formula; PivotTable + chart technique.



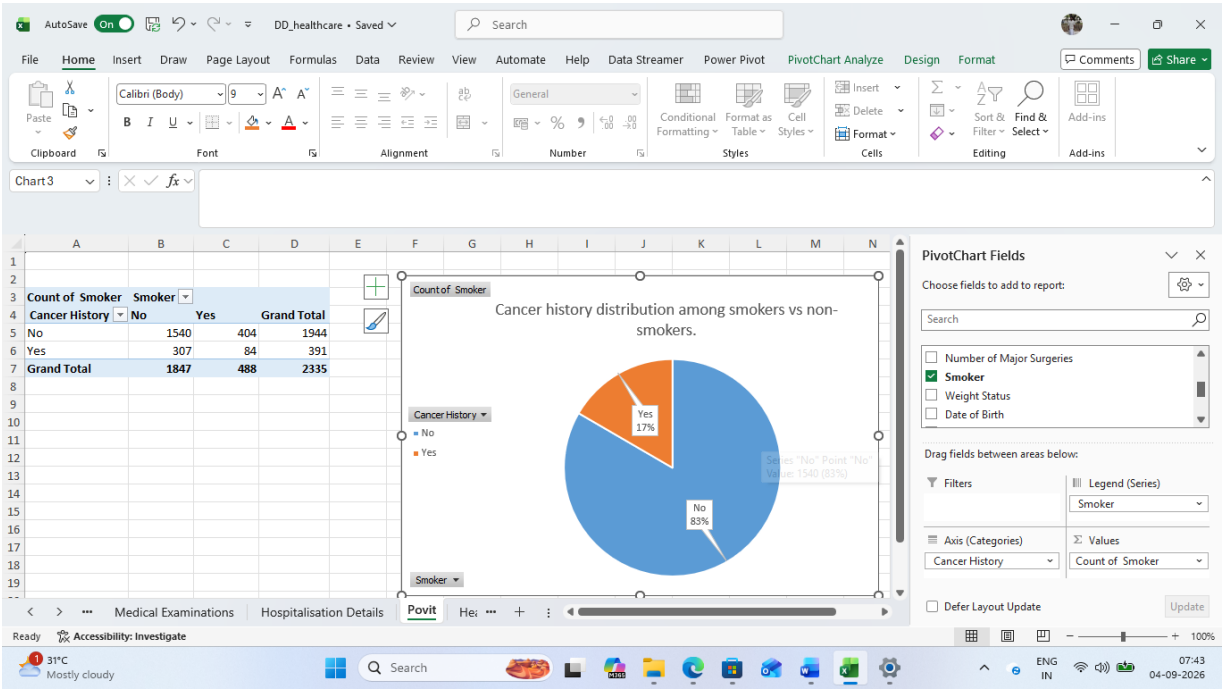
Activity 2: Manual Dashboard Build (should be done manually) – Insight

Question: Draw the insight manually and marked it down in the sheet

Explanation: This means write a short interpretation of what the displayed PivotTable/chart shows directly in the worksheet.

Technique used: Manual written insight beside or below the visualization.

Formula used: No formula; manual interpretation.



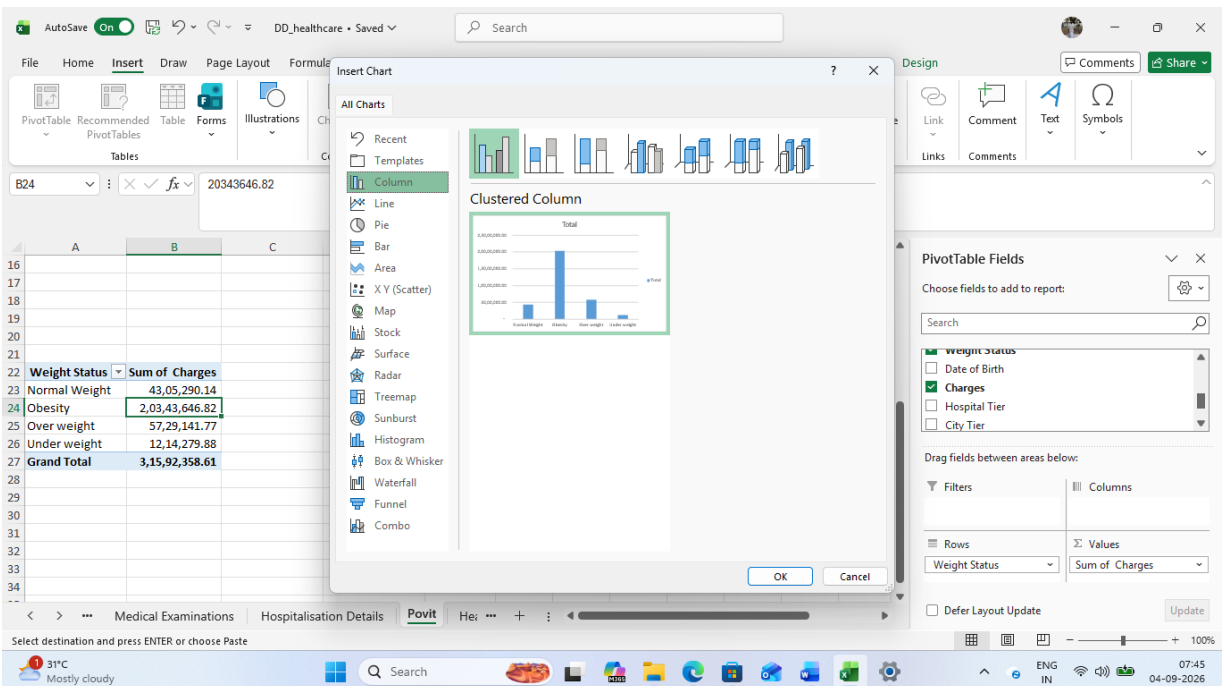
Activity 2: AI-Powered Summary Statistics & Visualization (AI-driven task)

Question: Automatically generate and visualize overall healthcare statistics.

Explanation: This means use Excel's Quick Analysis tools to generate summary statistics and visualizations from the combined Healthcare data.

Technique used: Quick Analysis / Recommended Charts.

Formula used: No single formula is specified; the source requires Quick Analysis.



AutoSave

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Search

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Paste

Clipboard

Calibri

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A^A

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Font

Alignment

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Conditional Formatting

Format as Table

Cell Styles

Styles

Insert

Delete

Format

Cells

Σ

Sort & Filter

Find & Select

Editing

Comments

Share

Add-ins

L2339

	A	B	C	D	E	F	G	H	I	J	K	L	M	N		
2319	d23118	Candice M	18.82	5.51	Yes	No	No		0	No	Normal Weight	18-09-1996	770.38	tier-3	tier-2	R101:
2320	d23119	Theodore A	17.08	5.73	No	No	Yes		1	No	Under weight	28-06-1993	770.00	tier-3	tier-3	R101:
2321	d23220	Lindy	19.21	5.53	Yes	No	No		0	No	Normal Weight	22-10-1996	760.00	tier-3	tier-3	R101:
2322	d23221	Thomas	17.86	5.43	No	No	Yes		1	No	Under weight	09-08-1993	760.00	tier-3	tier-1	R101:
2323	d23222	Holly	21.38	8.01	No	No	No		0	No	Normal Weight	19-09-2002	750.00	tier-3	tier-1	R101:
2324	d23223	Matthew	23.95	5.94	No	No	No		0	No	Normal Weight	14-12-1999	722.99	tier-3	tier-3	R101:
2325	d23224	Meghan K	22.24	5.04	No	No	No		0	No	Normal Weight	26-12-1999	700.00	tier-2	tier-3	R101:
2326	d23225	Fred	24.76	4.54	Yes	No	No		0	No	Normal Weight	12-09-2001	687.54	tier-3	tier-2	R101:
2327	d23226	Sebastian	20.1	5.6	Yes	No	Yes		1	No	Normal Weight	09-11-1997	670.00	tier-3	tier-3	R101:
2328	d23227	Laura	21.77	10.67	No	No	No		0	No	Normal Weight	29-11-2002	668.00	tier-3	tier-2	R101:
2329	d23228	Nicole	17.82	5.26	Yes	No	No		1	No	Under weight	04-07-1995	650.00	tier-3	tier-3	R101:
2330	d23229	Susan E	17.07	5.22	No	No	Yes		1	No	Under weight	01-06-1993	650.00	tier-3	tier-3	R101:
2331	d23300	Katy	22.24	4.29	Yes	No	No		0	No	Normal Weight	20-11-2001	644.14	tier-3	tier-3	R101:
2332	d23311	Jordan	22.34	5.57	No	No	No		1	No	Normal Weight	27-07-1998	637.26	tier-3	tier-3	R101:
2333	d23322	Juan D Sr	17.7	6.28	No	No	No		1	No	Under weight	13-09-1992	604.54	tier-3	tier-3	R101:
2334	d23333	Julie	16.47	6.35	No	No	Yes		1	No	Under weight	30-06-1993	600.00	tier-2	tier-1	R101:
2335	d23344	Evan P	17.6	4.39	No	No	No		1	No	Under weight	30-11-1992	570.62	tier-2	tier-1	R101:
2336	d23355	Aaron K	17.58	4.51	No	No	No		1	No	Under weight	09-07-1992	563.84	tier-2	tier-3	R101:
2337	2335											13529.918	2335			
2338																
2339																
2340																

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Activity 2 – Step 2

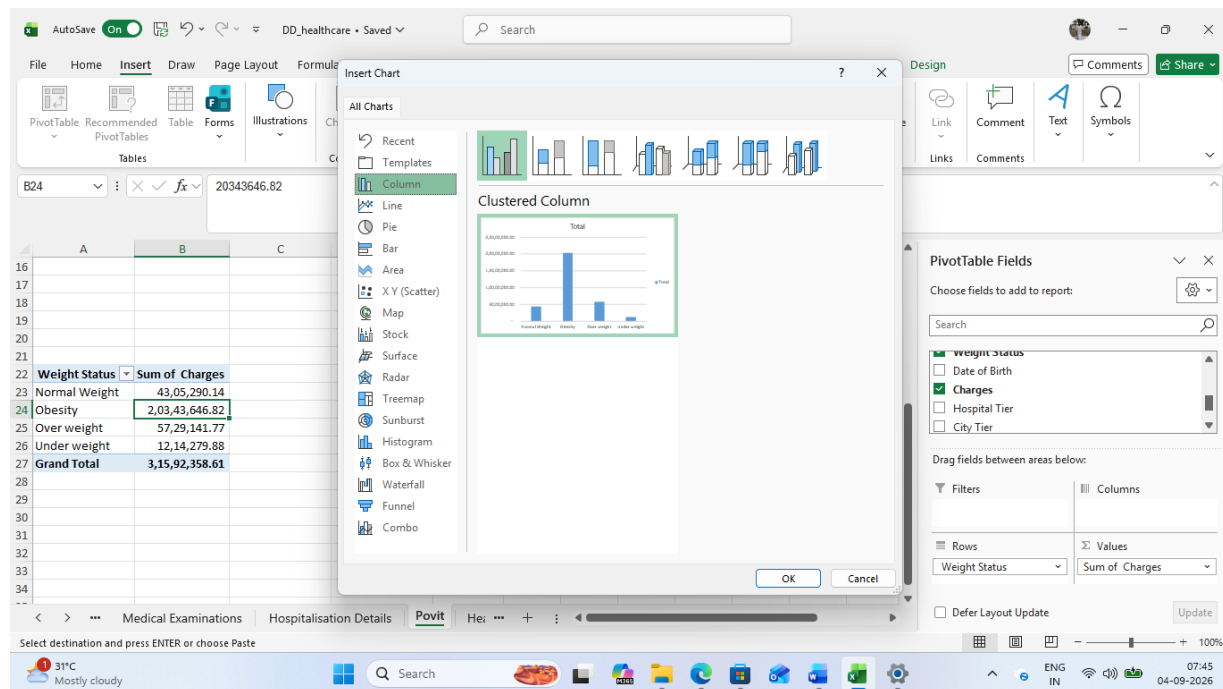
Question: Use Quick Analysis→ to calculate dynamic counts and averages →

- Total customers count,
- Total hospitalisation count,
- Total healthcare costs,
- Average costs and medical metrics

Explanation: This means generate the requested totals and averages directly from the combined dataset.

Technique used: Quick Analysis → Totals / summary calculations.

Formula used: No single formula is specified in the original question; use the Quick Analysis technique shown.



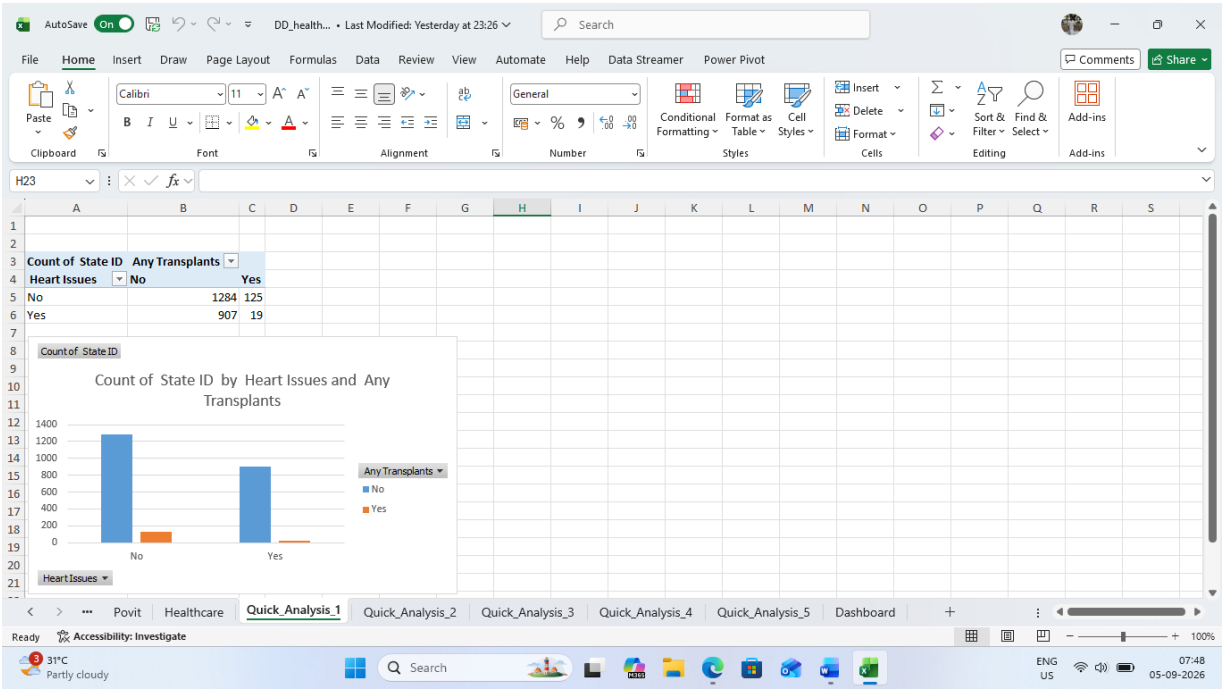
Activity 2 – Step 3

Question: Use Quick Analysis → Recommended Charts (Column/Bar/Doughnut) to visualise distributions of all pivots created in the previous step using AI tools.

Explanation: This means let Excel recommend suitable chart types for the summaries and use those charts to visualize the distributions.

Technique used: Quick Analysis → Recommended Charts.

Formula used: No single formula; chart recommendation technique.



Activity 2 – Expected Output

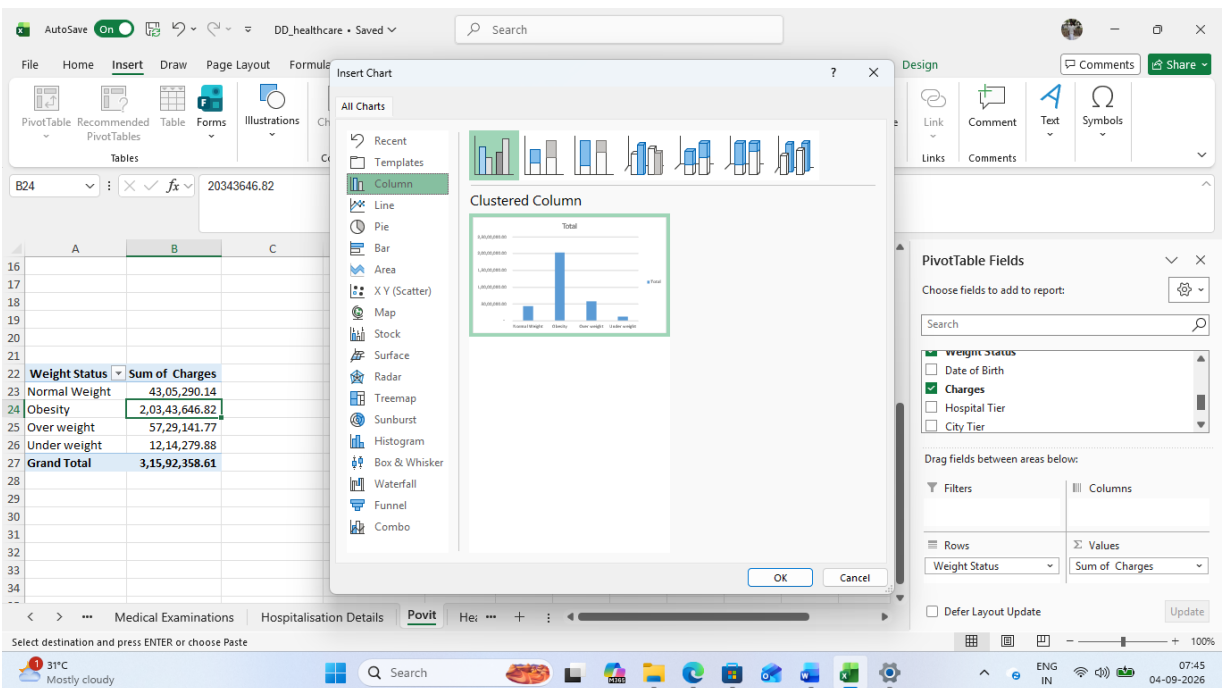
Question: Expected Output:

- Pivot chart/quick analysis chart visualising totals.

Explanation: This means the final output of this activity should include a chart that visually represents the calculated totals.

Technique used: PivotChart / Quick Analysis chart.

Formula used: No formula.



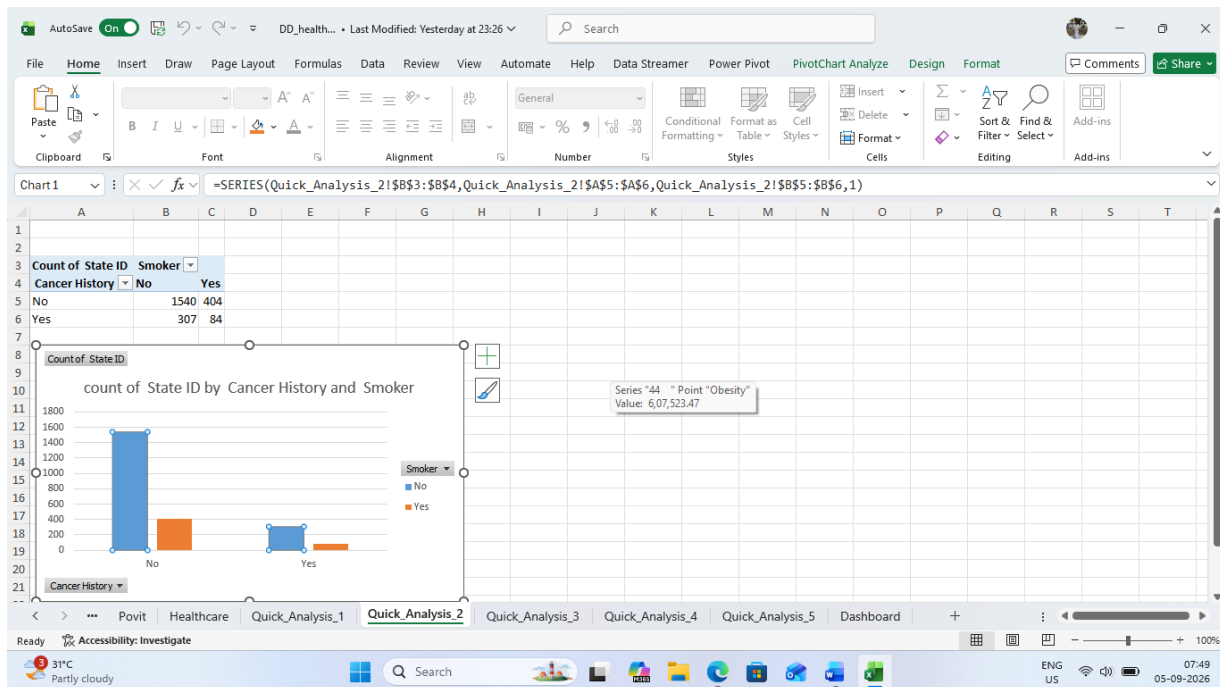
Activity 4: Dashboard Creation

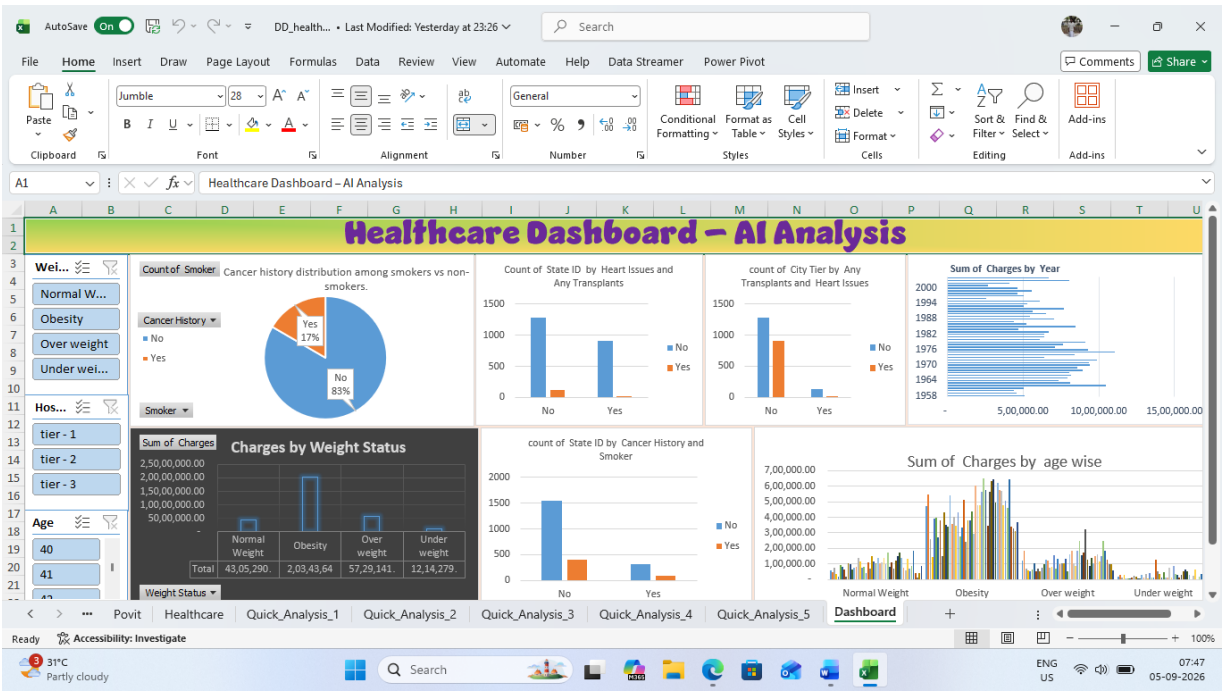
Question: Build an interactive dashboard that consolidates all key insights using the above visualisations. Ensure visual clarity and ease of interpretation for all chart types.

Explanation: This means bring the important charts together into one clear dashboard and make the layout easy to understand.

Technique used: Dashboard layout using PivotCharts/charts.

Formula used: No single formula; dashboard design technique.





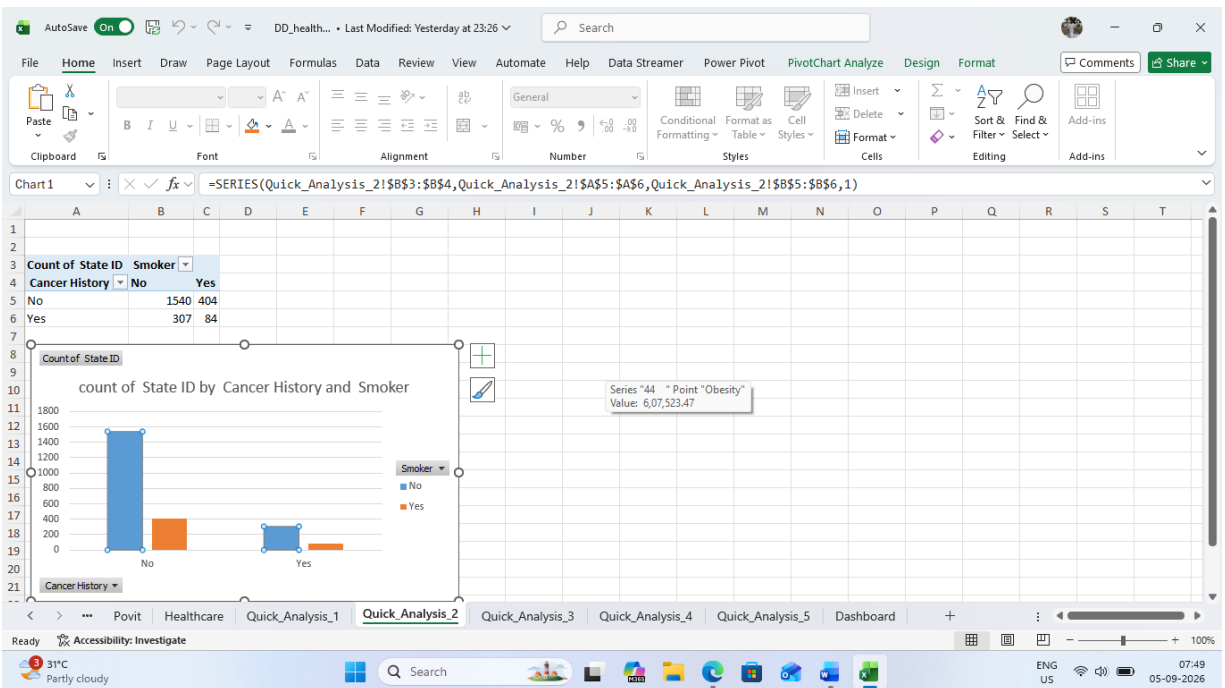
Activity 4: Dashboard Creation – Slicers

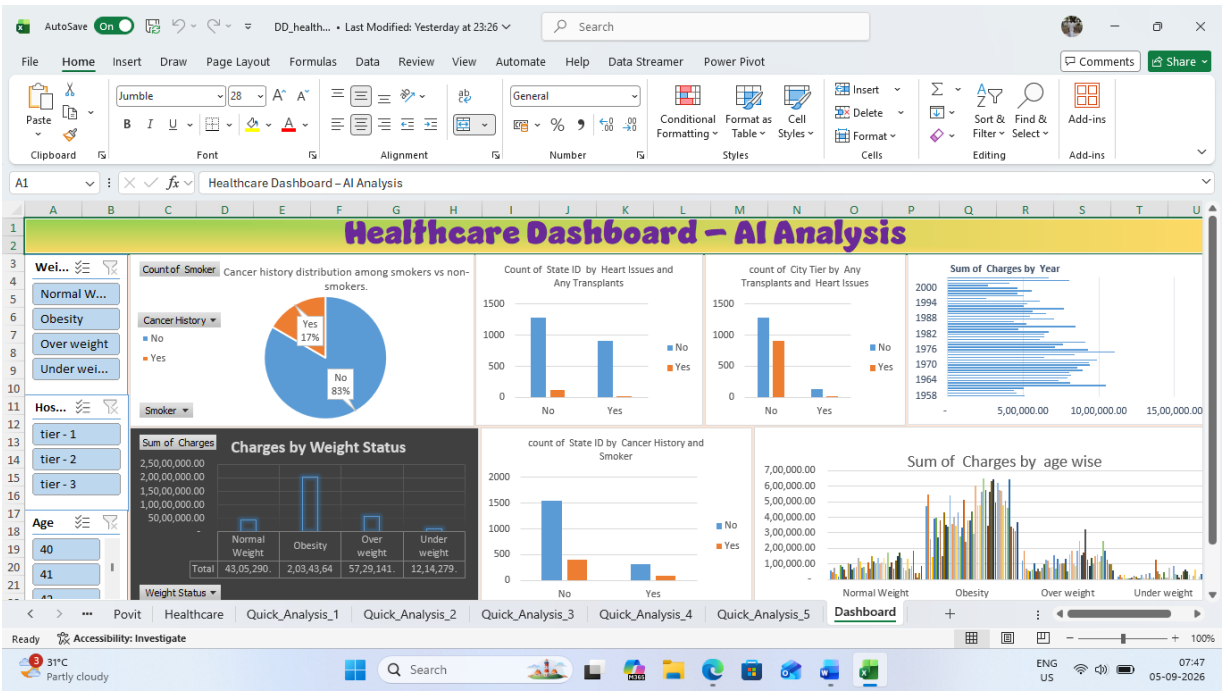
Question: Insert slicers for weight status and diabetes status, and wherever possible.

Explanation: This means add interactive filters so the dashboard can be filtered by Weight Status and Diabetes Status.

Technique used: Insert → Slicer using PivotTable/PivotChart connections.

Formula used: No formula; Slicer technique.





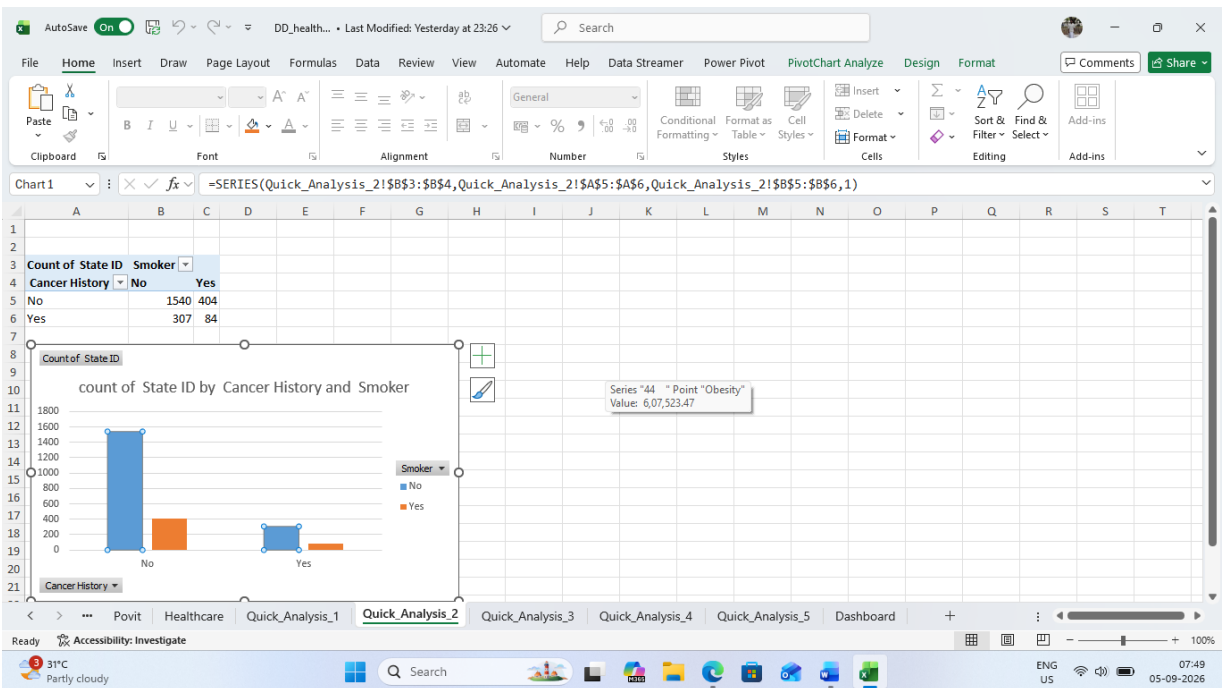
Activity 4: Dashboard Creation – Layout

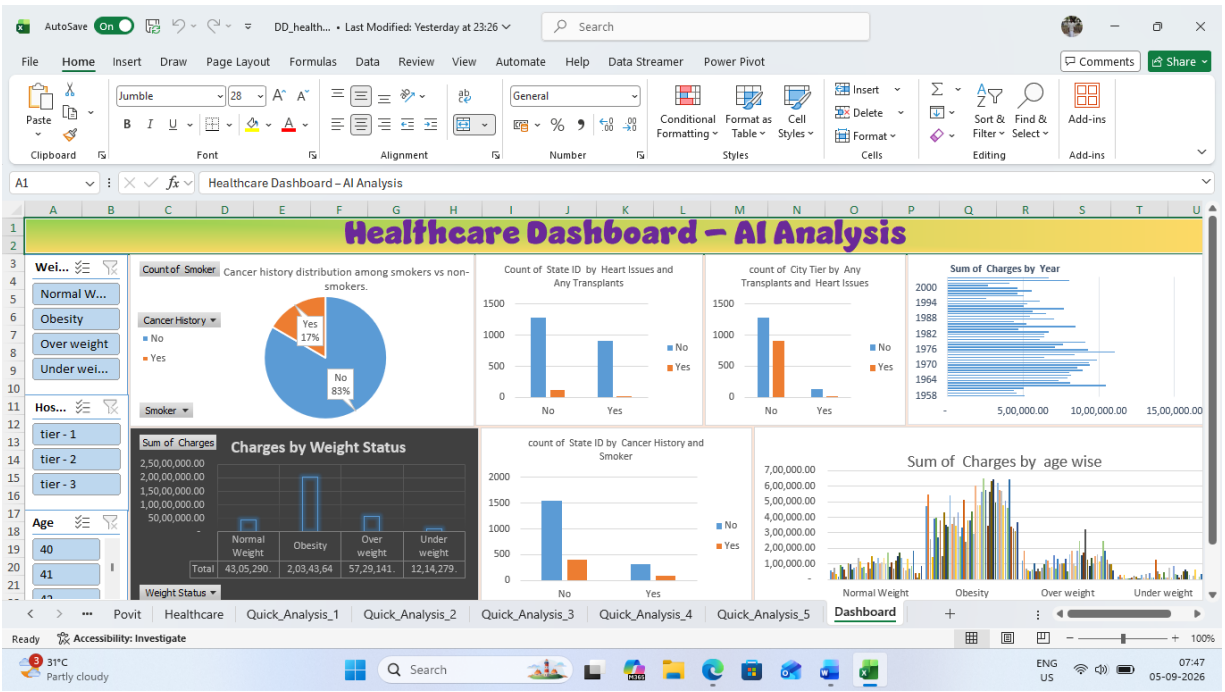
Question: Arrange charts and slicers into a simple dashboard layout

Explanation: This means position the charts and slicers neatly so users can read and interact with the dashboard easily.

Technique used: Manual dashboard arrangement and formatting.

Formula used: No formula.





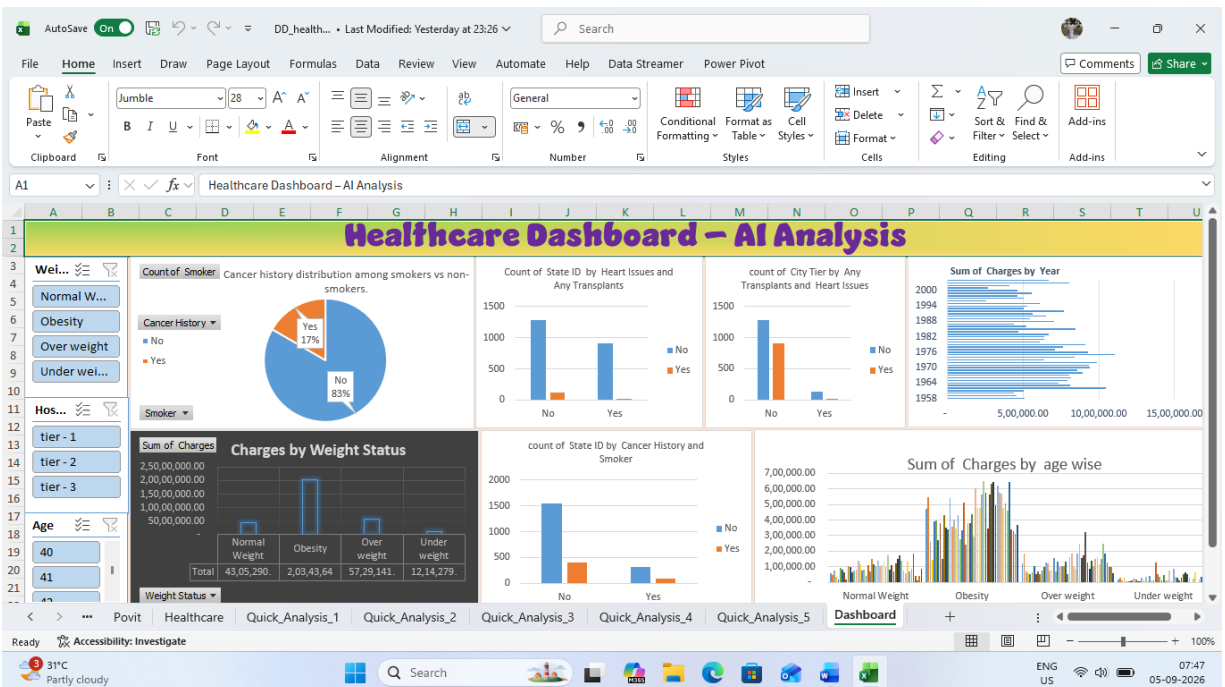
Activity 4: Dashboard Creation – Heading

Question: Add a heading (e.g., Healthcare Dashboard – AI Analysis).

Explanation: This means add a clear title at the top of the dashboard to identify the analysis.

Technique used: Cell formatting / merged or formatted heading area, as shown in the source screenshot.

Formula used: No formula.



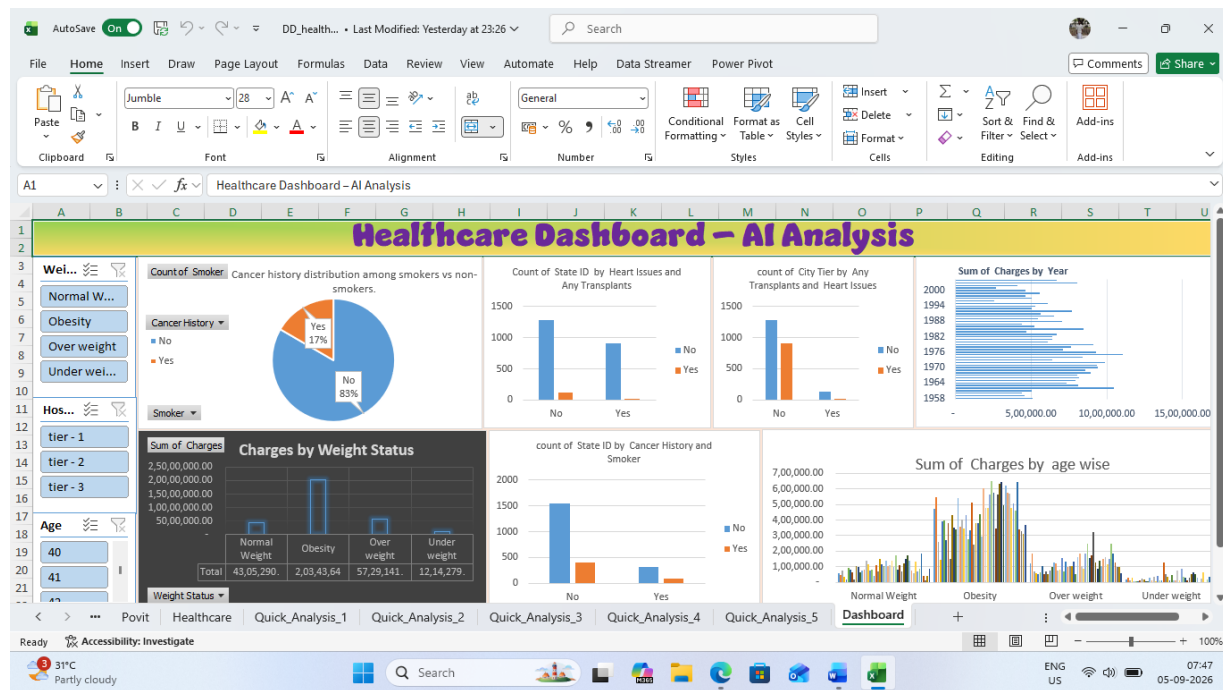
Activity 4: Dashboard Creation – Formatting

Question: Ensure consistency in chart formatting for clear interpretation.

Explanation: This means use consistent titles, chart styles, sizing and placement so the dashboard is easy to interpret.

Technique used: Manual chart formatting and dashboard design.

Formula used: No formula.



Final note: The original question wording is retained, and the formula/technique section distinguishes between formulas actually visible in the supplied screenshots and techniques required by the original question. No unsupported formula has been presented as if it came from the source.